

ABDN FELLOWSHIP TEACHING SERIES

MRI Physics for the Imaging Scientist

Magnetic Fields · Image Formation · Contrast Weightings
Pulse Sequences · fMRI Physics

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How to Use These Notes

These notes accompany the ABDN fellows' lecture on MRI physics. They are written for an audience that **uses** MRI data rather than **acquires** it: computer scientists training models, neuroscientists mapping brain circuits, anatomists measuring structures, engineers building reconstruction pipelines, and clinicians interpreting scans for research.

The philosophy is simple. You will probably never stand at the scanner console and choose a repetition time. But every dataset you will ever touch was shaped by someone who did — and the choices they made live inside your pixels. A model trained without knowing what TE does will fail on a new scanner and you will not know why. An fMRI result interpreted without knowing what T2* is may be an artefact rather than a finding. These notes exist to make you *literate in the language of the machine that generates your data*.

Each module follows the same rhythm: a plain-language picture first (an analogy you can repeat at a dinner table), then the physics stated precisely (with the one or two equations that genuinely matter), then worked examples and real clinical images, and finally a short *why you should care* box tied to research practice. Nothing here requires more than secondary-school mathematics to read; everything here is stated accurately enough to defend in a viva.

The five questions these notes answer.

1. What are the three magnetic fields inside the scanner, and what does each one do? (Module 1)
2. How does a signal from a spinning proton become a pixel with an address? (Module 2)
3. Why does the same brain look completely different on T1w, T2w, FLAIR, DWI and SWI? (Module 3)
4. What do the pulse-sequence acronyms (SE, TSE, GRE, EPI, IR) actually mean physically? (Module 4)
5. How can a scanner that sees water protons possibly image a thought? (Module 5)

MODULE 1

The MRI Machine and Its Magnetic Fields

Learning objectives

- Explain why MRI is built around hydrogen protons.
- Describe what B_0 does to proton populations and why it produces net magnetization.
- Define precession and compute the Larmor frequency at any field strength.
- Distinguish the roles of B_0 , B_1 and the three gradient fields.
- State the practical trade-offs of 1.5T vs 3T vs 7T.

1.1 What MRI actually measures

MRI does not photograph anatomy. It measures the magnetic behaviour of hydrogen protons — mostly the protons in water and fat — and reconstructs an image from that behaviour. The body is roughly 60–70% water, so every tissue carries an enormous, built-in signal source. No injected tracer, no ionizing radiation: the contrast comes from physics, not from a dye (though contrast agents can be added later).

Three ingredients do all the work: **a very strong static magnet** (B_0), **radio waves** (B_1 pulses), and **switchable gradient fields** (G_x , G_y , G_z). A computer orchestrates them and turns the returning radio signals into images. Everything in these five modules is a consequence of how those three magnetic fields interact with spinning protons.

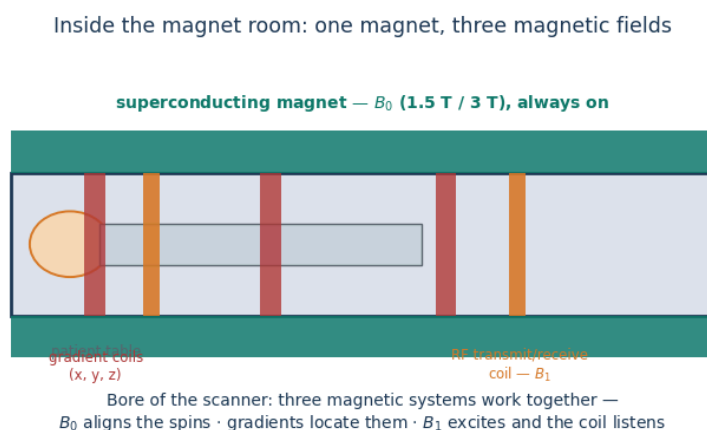


Figure 1.1. The bore of the scanner. Three magnetic systems surround the patient: the superconducting B_0 magnet (always on), the gradient coils (x , y , z), and the RF transmit/receive coil that produces B_1 and listens for the returning signal.

1.2 Protons: the tiny magnets inside us

The nucleus of hydrogen is a single proton. It has a positive charge and a quantum property called *spin*, and a moving charge generates a magnetic field — so each proton behaves like a microscopic bar magnet with a north and a south pole. Outside a magnetic field, billions of these tiny magnets point in random directions and their fields cancel: no net signal exists to measure.

Picture it. A stadium full of football fans, each waving a flag in a random direction. From the commentary box you see chaos, no pattern. The fans are your protons before the magnet is switched on. MRI begins the moment someone gives the crowd a reason to face the same way.

Why hydrogen and not another nucleus? Two reasons, both decisive. First, hydrogen is the most abundant nucleus in the body (water and fat are everywhere). Second, it has the largest gyromagnetic ratio of the

common biological nuclei ($\gamma = 42.58$ MHz/T, compared with 17.24 for ^{31}P , 11.26 for ^{23}Na and 10.71 for ^{13}C), so its signal is intrinsically loud. Abundant *and* loud: hydrogen wins on both counts.

1.3 B_0 : the main magnetic field and alignment

When a patient slides into the bore, B_0 — typically 1.5 or 3 tesla, roughly 30,000–60,000 times the Earth's magnetic field — acts on every proton at once. Protons settle into one of two energy states: **parallel** (with the field, lower energy) or **anti-parallel** (against it, higher energy). Slightly more choose parallel; at body temperature the excess is only a few protons per million, but across the trillions of protons in a voxel that small excess adds up to a measurable **net magnetization vector, M** , pointing along B_0 .

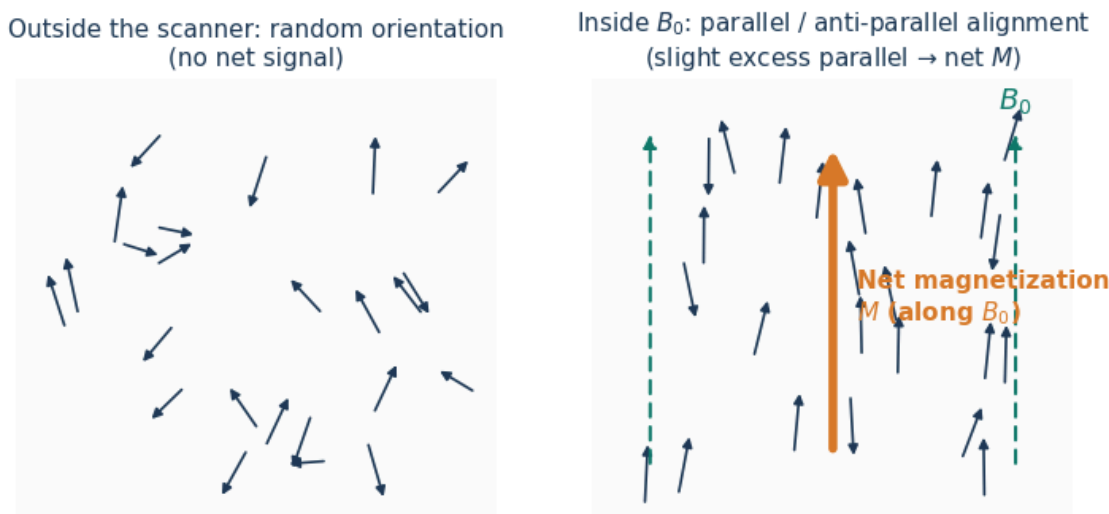


Figure 1.2. Left: outside the scanner, proton moments point randomly and cancel. Right: inside B_0 , protons align parallel or anti-parallel; the slight parallel excess creates the net magnetization M that all MRI signal ultimately comes from.

Why B_0 alone produces no image. M points along B_0 and just sits there. A receiver coil detects only *changing* magnetic flux (Faraday induction), and a static vector changes nothing. B_0 *prepares* the spins; it cannot *interrogate* them. For that we need B_1 — Module 1.5 — and to know *where* the signal came from we need the gradients — Module 1.6.

1.4 Precession and the Larmor frequency

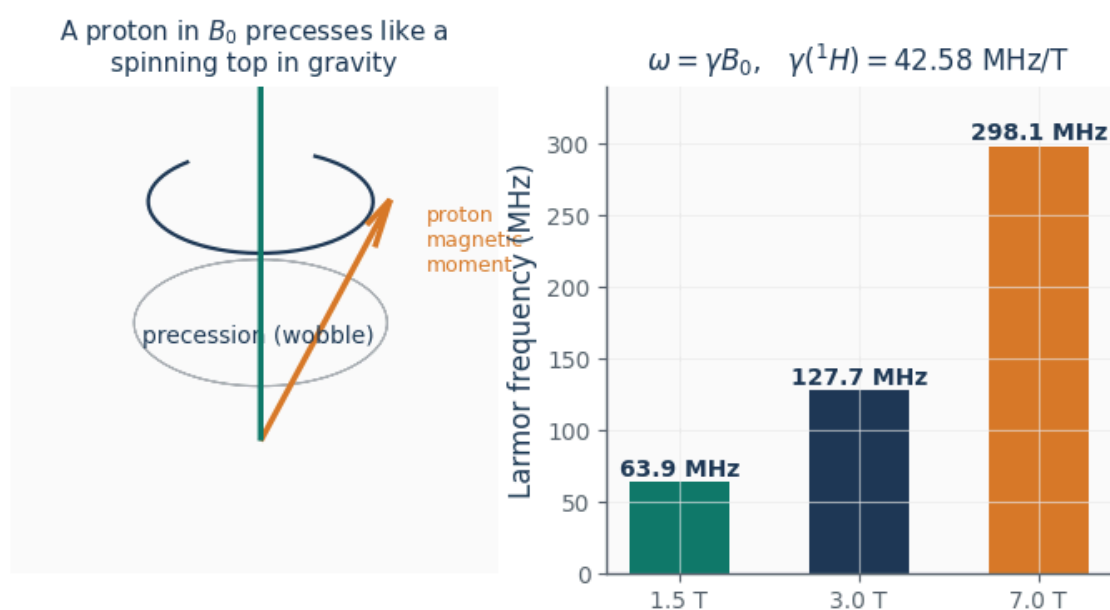
Aligned protons do not sit still. Because they are spinning, the torque from B_0 makes them **precess** — wobble around the field direction, exactly like a spinning top wobbling around gravity. The rate of that wobble is not arbitrary. It is set by one equation, the most important in all of MRI:

$$\omega = \gamma B_0 \quad (1.1)$$

where ω is the **Larmor (precession) frequency**, γ is the gyromagnetic ratio of the nucleus (42.58 MHz/T for hydrogen), and B_0 is the main field strength. Worked examples:

Table 1.1. Larmor frequency of hydrogen at common field strengths, from $\omega = \gamma B_0$

Field strength	Calculation	Larmor frequency	Typical use
1.5 T	42.58×1.5	≈ 63.9 MHz	Routine clinical imaging
3.0 T	42.58×3.0	≈ 127.7 MHz	High-resolution neuroimaging, fMRI
7.0 T	42.58×7.0	≈ 298.1 MHz	Research: cortical layers, spectroscopy

**Figure 1.3.** Left: a proton precesses around B_0 like a wobbling top. Right: the Larmor frequency scales linearly with field strength — protons at 3T precess exactly twice as fast as at 1.5T.

Picture it. Every nucleus is a radio station broadcasting at its own fixed frequency, and γ is what sets the dial. Hydrogen in a 1.5T scanner broadcasts at 63.9 MHz, always, predictably. To talk to the protons, the scanner must transmit on exactly their station. Slightly off-frequency, and the protons simply do not answer. That selectivity is called **resonance**, and it is why the technique is called magnetic resonance imaging.

1.5 B_1 : the radiofrequency field and resonance

B_1 is a second, much weaker magnetic field, switched on in brief bursts by the RF coil and oscillating at the Larmor frequency. Because it matches the protons' precession frequency, energy transfers efficiently (resonance) and the net magnetization tips away from B_0 by an angle called the **flip angle** (α). A 90° pulse tips M fully into the transverse (xy) plane; smaller angles are used in fast imaging. When B_1 switches off, M is left rotating in the transverse plane — and a rotating magnetic vector is precisely what a receiver coil can detect. The B_1 pulse is therefore the moment the invisible becomes measurable.

Picture it. B_0 is the parade-ground general who keeps the soldiers standing at attention. B_1 is the visiting officer who arrives, shouts one sharp order — "turn!" — and leaves. The soldiers (protons) break ranks together, in step, and it is that synchronized movement the scanner records. When the order fades, they drift back to attention — and how fast each group drifts back is the contrast you will meet in Module 2.

1.6 The gradients: G_x , G_y , G_z

Gradient coils are electromagnets that add a small, controlled, *position-dependent* tilt to the main field. With a gradient on, the field — and therefore the Larmor frequency — varies smoothly from one side of the patient to the other. Since frequency and phase now encode position, the scanner can work out where every signal originated. There are three sets of gradient coils, one per axis, and each has a specific job in image formation:

Table 1.2. The three gradient axes and their roles (detailed in Module 2.3)

Gradient	Axis	Job	When it acts
G_z — slice select	z	Only one slice resonates with the RF pulse	During RF excitation
G_y — phase encode	y	Gives each row a distinct phase twist	Briefly before readout
G_x — frequency encode (readout)	x	Gives each column a distinct frequency	During signal collection

Picture it. Gradients are GPS for protons. B_0 gets everyone facing north; B_1 makes everyone sing; the gradients make each row of the choir sing at a different pitch and each column start on a different beat. Listen to the chord and you can tell exactly who sang — and where they were standing.

1.7 Field strength: 1.5T, 3T, 7T and safety

Doubling B_0 does not simply "double the image quality." The gains and the costs scale together:

Table 1.3. What changes as field strength increases

Property	Higher field →	Practical consequence
Signal (SNR)	Rises (roughly linearly)	Finer resolution or faster scans
T1 of tissues	Lengthens	Contrast behaviour changes; protocols must be re-tuned
Susceptibility effects	Stronger	More distortion near air/bone/implants — but better SWI and BOLD
RF energy deposition (SAR)	Rises steeply ($\approx B_0^2$)	Heating limits bind harder at 3T and above

B ₁ uniformity	Worsens	Shading artefacts, especially at 7T
Chemical shift	Larger	Fat/water misregistration increases

For a researcher pooling data across sites, the message of Table 1.3 is that **field strength is a study variable, not a scanner spec**. A 1.5T image and a 3T image of the same brain are not interchangeable inputs for an analysis pipeline or a trained model — their contrast, resolution and artefact profiles differ systematically.

Module 1 take-home. B₀ aligns the spins and sets their precession frequency ($\omega = \gamma B_0$). B₁, tuned to exactly that frequency, tips them into the measurable plane. The gradients stamp each signal with a positional address. Every image, artefact and contrast you will ever meet is these three fields, choreographed in time.

MODULE 2

Image Formation: From Spins to Pixels

Learning objectives

- Explain what a 90° RF pulse makes measurable, and what the flip angle controls.
- Define T1, T2 and T2* precisely and state their typical tissue values.
- Describe how slice selection, phase encoding and frequency encoding localize signal.
- Explain what k-space is, what its centre and periphery contain, and how the image is recovered from it.

2.1 Tipping the magnetization: the flip angle

At equilibrium M lies along B₀, invisible to the coil. A 90° B₁ pulse rotates it fully into the transverse plane, where it precesses at the Larmor frequency and induces an oscillating voltage — the MR signal. The flip angle is a genuine design choice: a full 90° gives maximum signal per excitation, while smaller angles (10–40°) let you repeat excitations rapidly without waiting for full recovery, which is how fast gradient-echo imaging works. Nothing about the anatomy changed; only the geometry of the magnetization changed — and that alone created the signal.

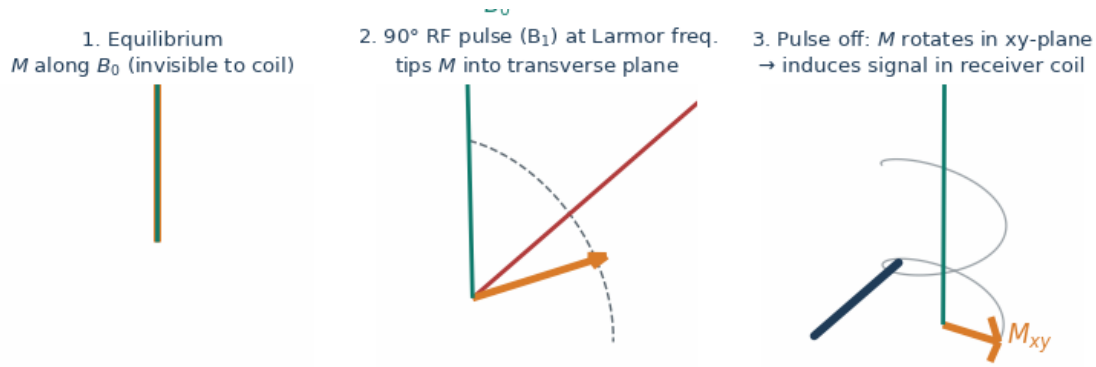


Figure 2.1. The three moments of excitation: M aligned with B_0 (undetectable), the 90° B_1 pulse tipping it into the transverse plane, and the rotating transverse magnetization inducing signal in the receiver coil.

2.2 Relaxation: T1, T2 and T2*

The instant the RF pulse stops, two independent processes begin *simultaneously*. They are the origin of all MRI contrast, so they deserve careful definitions.

T1 — spin–lattice relaxation (recovery)

T1 describes how fast M regrows along B_0 as protons hand their excitation energy to the surrounding tissue (the "lattice"). It is an exponential recovery:

$$M_z(t) = M_0 \left(1 - e^{-t/T_1} \right) \quad (2.1)$$

After one T1, 63% of the magnetization has recovered. T1 is short where molecular motion is well matched to the Larmor frequency (fat, ≈ 260 ms at 3T) and long where water tumbles too fast to exchange energy efficiently (CSF, ≈ 4000 ms). T1 lengthens with field strength — one of the reasons 1.5T and 3T images of the same tissue look different.

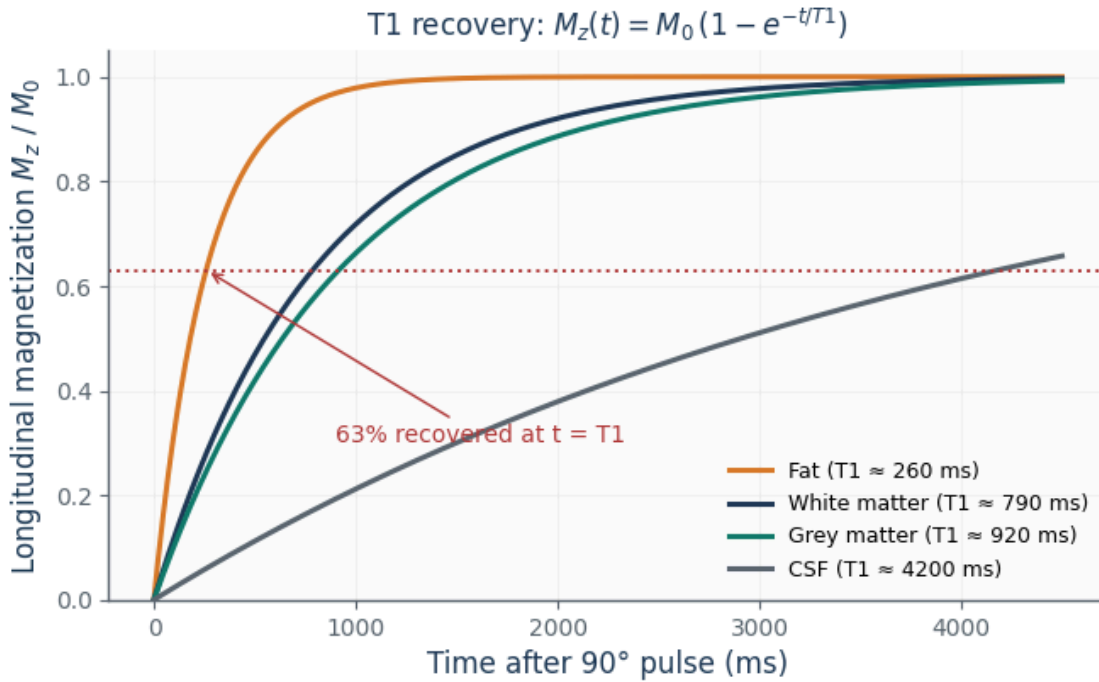


Figure 2.2. T1 recovery for four tissues (values at 3T). Fat recovers quickly; CSF barely recovers within seconds. At a short repetition time, fat has regrown most of its signal and CSF almost none — that difference is T1 contrast.

T2 — spin–spin relaxation (decay)

T2 describes how fast the transverse signal dies as protons exchange energy with *each other* and drift out of phase. No energy is lost to the lattice; coherence is lost among neighbours:

$$M_{xy}(t) = M_0 e^{-t/T_2} \quad (2.2)$$

After one T2, only 37% of the signal remains. CSF keeps phase coherence for a long time ($T_2 \approx 2000$ ms), so it is still "singing in unison" long after muscle ($T_2 \approx 45$ ms) has gone silent. Dephasing is always faster than energy recovery, so in every tissue $T_2 \leq T_1$ — usually much less.

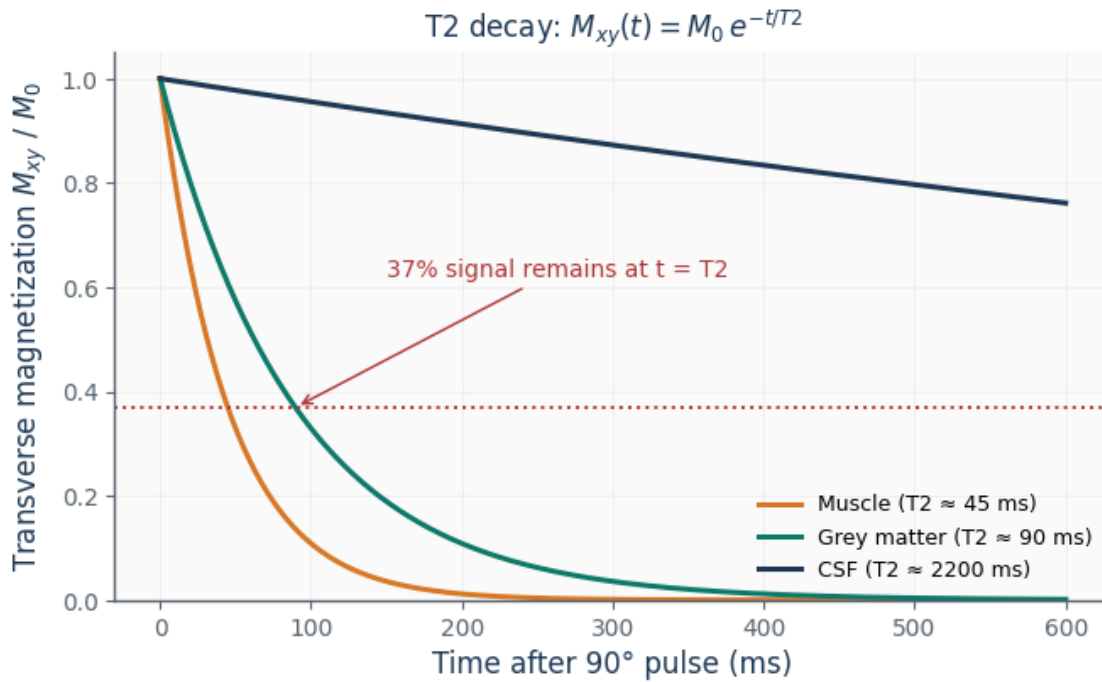


Figure 2.3. T2 decay. At a late echo time, long-T2 fluid still has signal while short-T2 muscle has decayed — that difference is T2 contrast.

T2* — the honest decay (tissue + machine)

In a real scanner the main field is never perfectly uniform, and every air–tissue or bone–tissue interface adds its own local distortion. These extra dephasing sources accelerate the decay beyond true T2:

$$\frac{1}{T2^*} = \frac{1}{T2} + \frac{1}{T2'} \quad (2.3)$$

where $T2'$ captures everything the tissue is not responsible for — field inhomogeneity, susceptibility interfaces, iron, blood products. $T2^*$ is always shorter than $T2$. Two things you will meet constantly depend on $T2^*$: gradient-echo imaging (Module 4.3), which cannot rewind $T2'$ effects, and BOLD fMRI (Module 5), which *exploits* them.

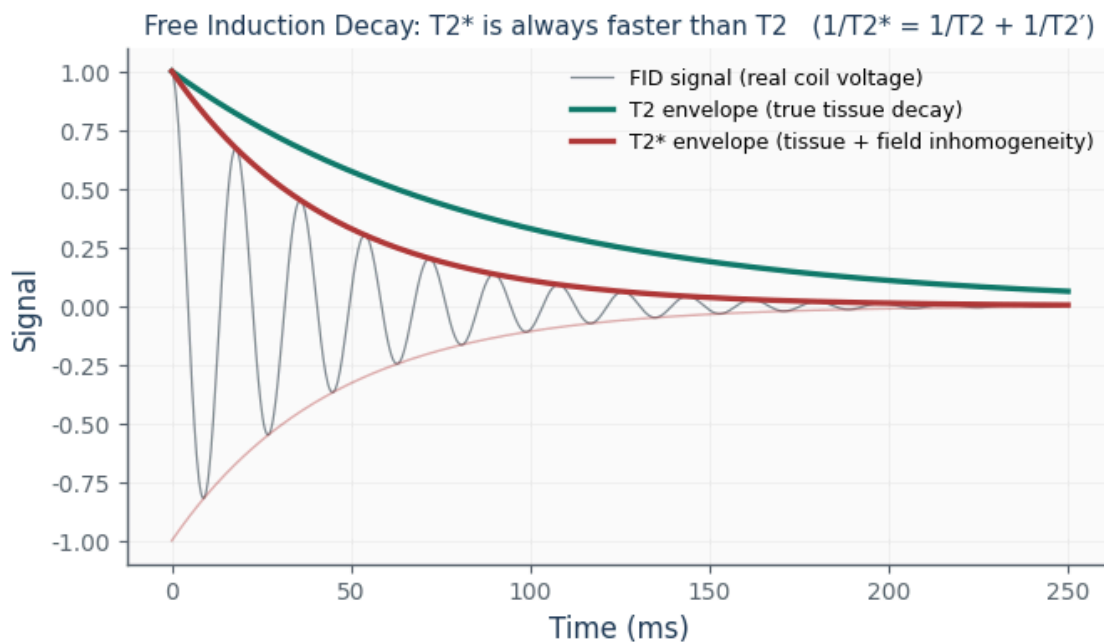


Figure 2.4. The free induction decay. The oscillating grey line is what the coil actually measures; it dies inside the T_2^* envelope, which always decays faster than the true tissue T_2 envelope.

Table 2.1. Representative relaxation times in brain at 3T (approximate; values vary by tissue state and field strength)

Tissue	T1 (ms)	T2 (ms)	Bright on T1w?	Bright on T2w?
Fat / subcutaneous	≈ 260	≈ 80	Yes	Intermediate
White matter	≈ 790	≈ 80	Brighter than grey	Darker than grey
Grey matter	≈ 920	≈ 90	Intermediate	Intermediate
CSF	≈ 4200	≈ 2200	No (dark)	Yes (very bright)
Acute haematoma (deoxy-Hb)	$\approx$ isointense	short	No	No (dark)

2.3 Spatial encoding with gradients

Relaxation tells us *what* the tissue is doing; it says nothing about *where*. Localization is the gradients' job, and it happens in three coordinated steps, each exploiting $\omega = \gamma B$.

Step 1 — Slice selection (G_z)

Switch G_z on during the RF pulse. Now the Larmor frequency varies linearly along z , so a B_1 pulse with a narrow bandwidth excites *only* the thin slab whose frequency matches. Everyone outside the slice simply does not resonate. Slice position is chosen by the RF centre frequency; slice thickness by the bandwidth and gradient strength.

Step 2 — Phase encoding (G_e)

Apply G_e briefly, then switch it off. While it was on, spins at different y-positions precessed at different speeds; when it stops they return to a common frequency but *remember* how far ahead or behind they fell — as a position-dependent phase. This step encodes only one line of data, so it must be repeated with many different G_e strengths. Phase encoding is why scans take minutes, not seconds.

Step 3 — Frequency encoding (G_x)

During signal collection, G_x makes precession frequency vary along x. The measured signal is a chord of many frequencies; a Fourier transform separates them, and each frequency maps to an x-position. The pitch tells you the column; the phase memory tells you the row.

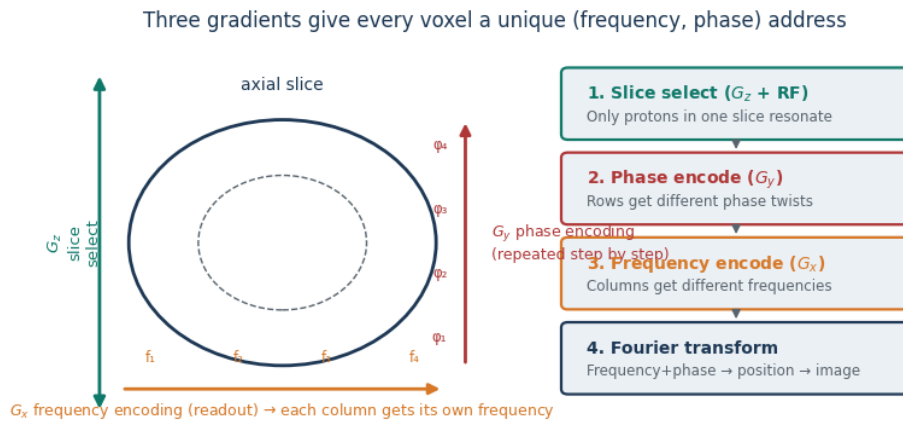


Figure 2.5. The three-step localization scheme: G_z selects the slice, G_e stamps each row with a phase, G_x stamps each column with a frequency, and the Fourier transform turns the coded signal back into positions.

Why this matters for your research. Slice thickness and slice gap are acquisition choices, not anatomy. A 4 mm lesion in a 5 mm slice with a 1 mm gap can vanish into *partial volume averaging*. In-plane resolution and through-plane resolution are different numbers with different consequences — check both before trusting a measurement taken from someone else's scan.

2.4 k-space: the raw data of MRI

The scanner never stores an image directly. It stores a matrix of spatial-frequency measurements called **k-space**: each data point contains signal from the *whole* slice, encoded at one particular combination of phase and frequency. The image and k-space are a Fourier pair — one is the 2D Fourier transform of the other:

$$I(x, y) = \mathcal{F}^{-1} \{S(k_x, k_y)\} \quad (2.4)$$

The geography of k-space is one of the most useful facts in practical MRI:

- **Centre (low spatial frequencies):** carries the bulk of the signal — overall brightness and **contrast**. Lose the centre and the image is a faint outline.
- **Periphery (high spatial frequencies):** carries edges and fine detail — **resolution**. Lose the periphery and the image is a blurry blob with correct contrast.

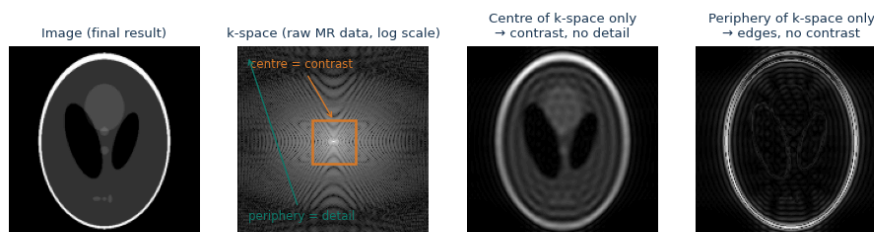


Figure 2.6. A real Fourier demonstration on a standard test phantom. Reconstructing from the centre of k-space alone preserves contrast but destroys detail; reconstructing from the periphery alone preserves edges but destroys contrast. The full image needs both.

Three consequences follow immediately. First, **motion corrupts k-space, not pixels**: a patient who moves between two phase-encoding steps creates structured ghosting across the whole image — which is why motion correction is a Fourier-domain problem, not a "sharpen the blurry photo" problem. Second, you can **skip lines** (partial Fourier, parallel imaging, compressed sensing) to go faster, paying with SNR or reconstruction artefacts — the entire field of deep-learning reconstruction lives in this trade. Third, sequence families differ in *how* they walk through k-space, which is exactly the subject of Module 4.

Module 2 take-home. RF tips M into the measurable plane; T1 recovery and T2/T2* decay give each tissue a signature; the gradients encode position as frequency and phase; k-space accumulates the encoded signal; the inverse FFT turns it into the image on your screen. Pixels are the end of the story, not the beginning.

MODULE 3

Contrast Weightings: Reading the Images

Learning objectives

- Define TR and TE and predict contrast from their values.
- Recognize T1w, T2w, PD, FLAIR, STIR, DWI/ADC and SWI/T2*w images at a glance.
- State what each weighting is for, and what can go wrong when it is misread.

3.1 TR and TE: the two dials of contrast

Every MRI sequence repeats a basic cycle: excite, encode, listen, repeat. Two timing parameters dominate how the result looks:

- **TR (repetition time):** how long the scanner waits between excitations. A short TR catches tissues mid-recovery, so their *T1 differences* show. A long TR lets everyone recover, erasing T1 differences.
- **TE (echo time):** how long the scanner waits after excitation before listening. A long TE lets fast-decaying tissues go quiet first, so their *T2 differences* show. A short TE listens before decay separates anyone.

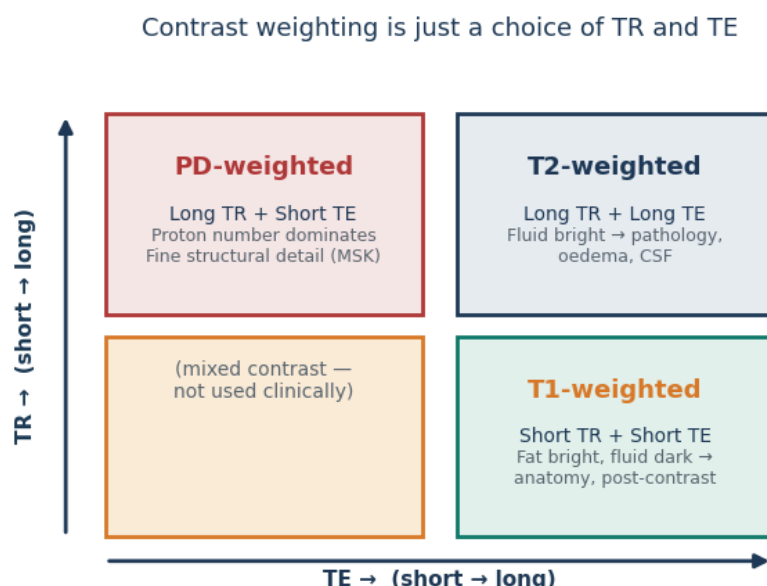


Figure 3.1. The contrast matrix. Choose TR and TE and you have chosen your weighting: short/short = T1w, long/long = T2w, long/short = PD. "Weighting" simply means *which tissue property dominates the image*.

Picture it. TR is how long you let the choir rest between songs; TE is how long you wait before judging who is still singing. Judge too soon (short TE) and everyone sounds equal. Let them rest fully (long TR) and the T1 differences vanish. The conductor's two choices decide which choir members stand out — and the radiographer's two choices decide which tissues stand out.

3.2 T1w, T2w and PD

Table 3.1. The three fundamental weightings

Weighting	TR	TE	Fluid (CSF)	Fat	Main use
T1-weighted	short (<1000 ms)	short (<30 ms)	dark	bright	Anatomy, grey/white differentiation, post-contrast imaging
T2-weighted	long (>2000 ms)	long (>80 ms)	bright	intermediate	Pathology: oedema, inflammation, most lesions

The mnemonic worth memorizing: **T1 = fat bright (anatomy)**; **T2 = water bright (pathology)**. Pathology almost always involves extra water — oedema, inflammation, demyelination, many tumours — so T2w is the workhorse "where is the disease?" sequence, while T1w is the "where exactly is the anatomy?" sequence, and the pair together triangulates most findings.

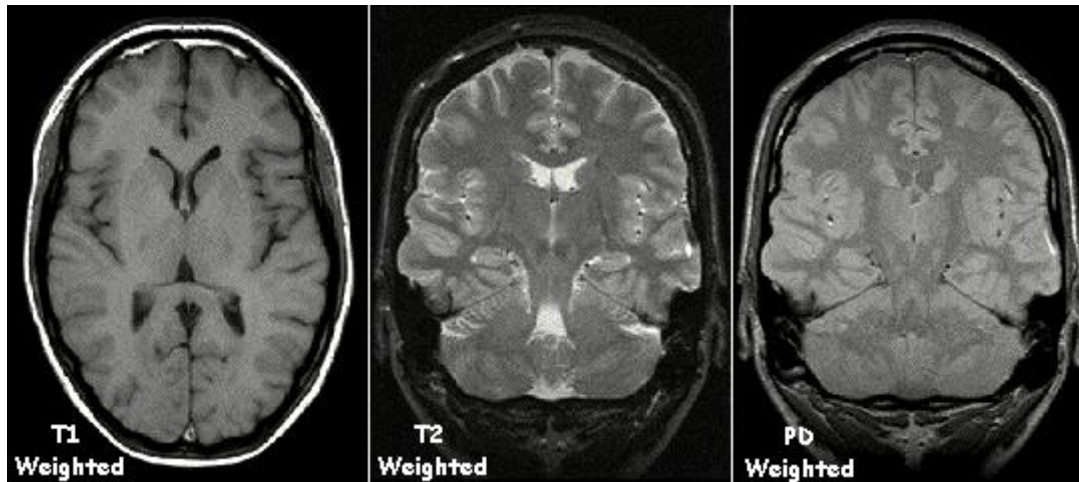


Figure 3.2. The same brain imaged three ways. On T1w, CSF in the ventricles is black and white matter is brighter than grey; on T2w the CSF is brilliant and the grey/white relationship inverts; PD sits between the two. Nothing about the patient changed — only TR and TE. Image: Wikimedia Commons, "T1t2PD.jpg".

Reading habit. Find the CSF first. If the ventricles are black, you are looking at T1w (or FLAIR); if they are white, T2w; if grey, PD. This two-second check orients you before you read a single finding — and it is also a sanity check on whether a dataset's labels match its files.

3.3 FLAIR and STIR: making a tissue disappear

Sometimes one tissue is so bright it hides its neighbours. Periventricular MS plaques sit beside brilliantly bright CSF on T2w; bone-marrow oedema hides inside bright fat. The solution is an **inversion recovery** preparation (physics in Module 4.5): invert all magnetization with a 180° pulse, then wait exactly until the offending tissue's magnetization crosses zero as it recovers — and image at that instant. That tissue contributes no signal and vanishes:

- **FLAIR** (fluid-attenuated inversion recovery): TI chosen to null CSF (≈ 2900 ms at 3T). CSF goes black, so lesions next to ventricles and sulci finally stand out. The single most important neuro sequence after T2w itself.
- **STIR** (short-tau inversion recovery): TI chosen to null fat (≈ 180 ms at 3T). Fat goes black, so oedema and marrow lesions shine through. Works at any field strength, unlike frequency-selective fat saturation.

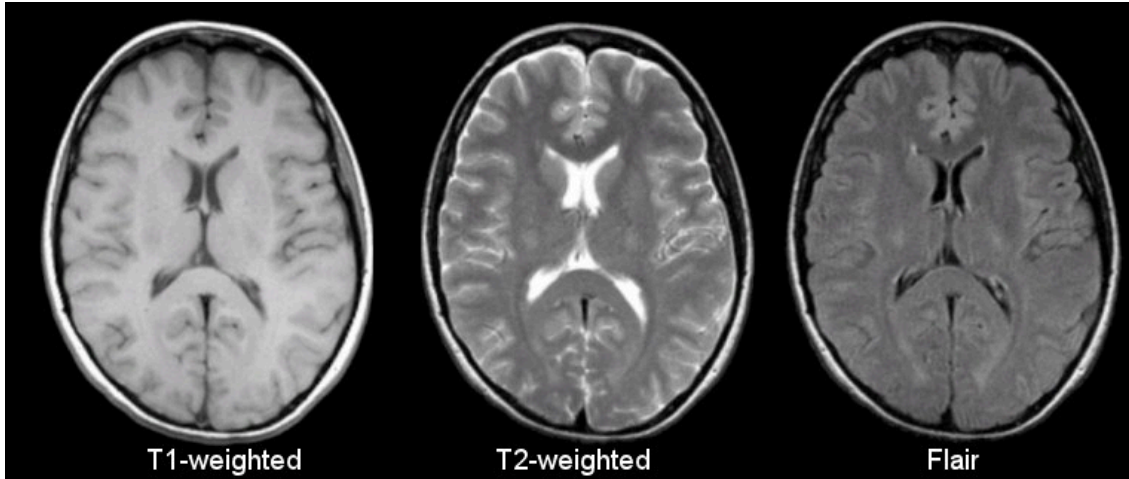


Figure 3.3. T1w, T2w and FLAIR of the same slice. FLAIR is a T2w image with the CSF signal switched off: brain lesions keep their T2 brightness while the ventricles go dark. Image: Case Western Reserve University, MRI Basics (Neurology).

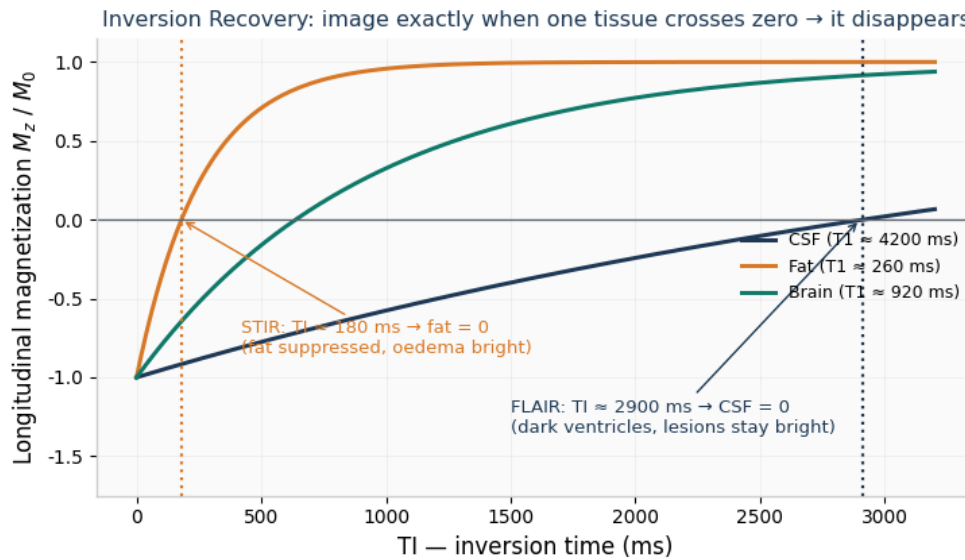


Figure 3.4. The physics behind Figure 3.3: after a 180° inversion pulse each tissue's magnetization recovers through zero at its own time ($TI \approx T1 \times \ln 2$). Image at fat's zero-crossing and you have STIR; image at CSF's zero-crossing and you have FLAIR.

3.4 DWI and ADC: imaging water motion

Diffusion-weighted imaging adds a pair of strong, matched gradients whose combined effect is to make signal sensitive to random microscopic water motion (Brownian motion). Water that moves freely between the two gradients loses phase coherence and goes dark; water that is trapped — inside swollen cells, in viscous pus, in densely packed tumour — keeps its signal and stays bright. The strength of the diffusion weighting is the **b-value** (s/mm^2); routine brain DWI uses $b \approx 0$ and $b \approx 1000$.

$$S(b) = S_0 e^{-b \cdot ADC} \quad (3.1)$$

Acquire at two or more b-values and equation (3.1) can be solved voxel-by-voxel for the **apparent diffusion coefficient (ADC)** — a quantitative map of how freely water actually moves. This matters because DWI brightness alone is ambiguous: DWI is a T2-weighted image with diffusion gradients bolted on, so a bright lesion might be truly restricted water *or* simply long-T2 fluid ("T2 shine-through"). The ADC map settles it:

The rule of DWI interpretation. DWI bright *and* ADC dark = true restricted diffusion (acute infarct, abscess, highly cellular tumour). DWI bright *but* ADC bright = T2 shine-through, an artefact, not restriction. Never report DWI without checking ADC.

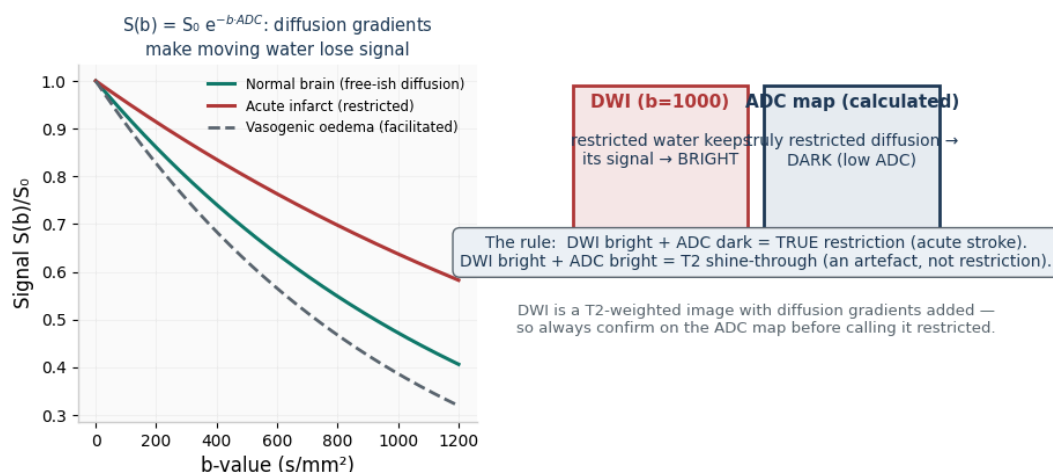


Figure 3.5. Left: signal falls with b-value according to each tissue's ADC — restricted water (red) keeps signal at high b. Right: the interpretation rule linking DWI and the ADC map.

3.5 SWI and T2*w: imaging field disturbances

Blood degradation products (deoxyhaemoglobin, methaemoglobin, haemosiderin), iron deposits and calcification are all magnetically different from brain tissue. Each one creates a small local field disturbance that accelerates T2* decay in its neighbourhood — and a T2*-weighted gradient-echo sequence turns those disturbances into signal voids. **Susceptibility-weighted imaging (SWI)** pushes this furthest: a high-resolution 3D T2*w acquisition whose phase information is used to amplify the contrast, making microbleeds, venous structures and iron visible as striking black foci.

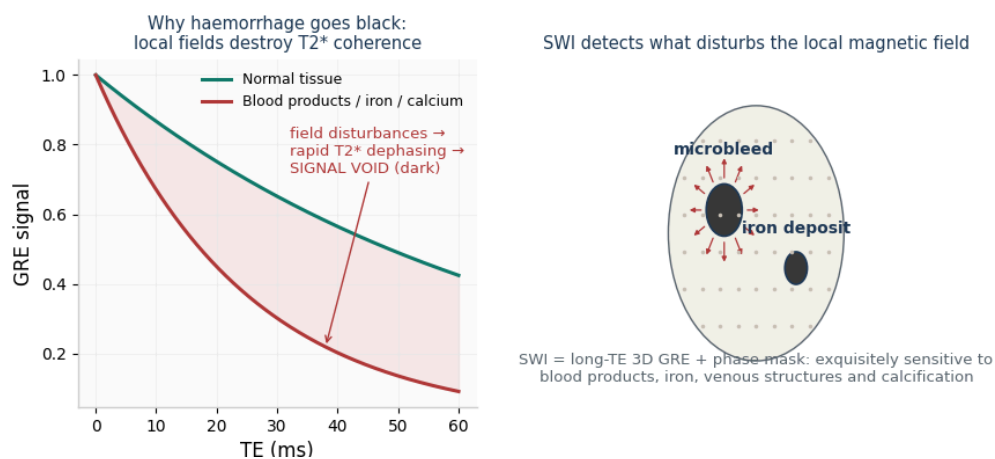


Figure 3.7. Why haemorrhage goes black on T2*w/SWI: local field disturbances destroy phase coherence rapidly, hollowing out the signal around blood products and iron.

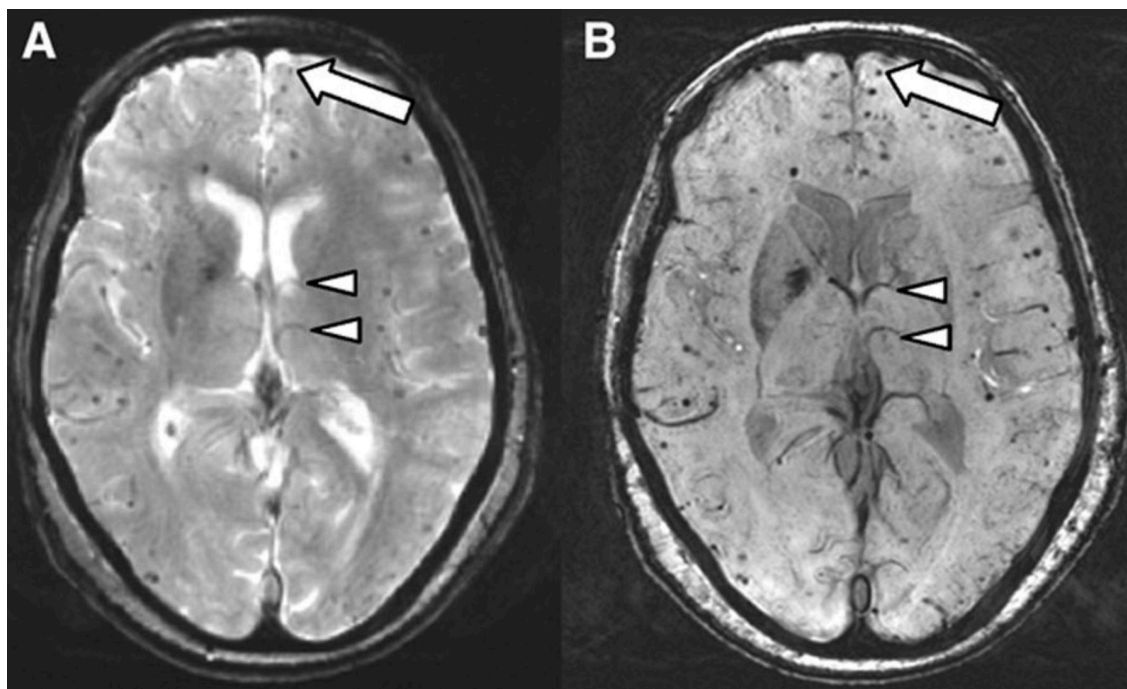


Figure 3.8. The same patient on conventional T2*w GRE (A) and SWI (B): SWI reveals dramatically more microbleeds (black dots) and delineates the basal-ganglia haemorrhage more clearly — the clinical value of engineering a sequence around T2*. Image: *Stroke* (American Heart Association Journals), SWI vs T2*w GRE for microbleed detection.

T1w	Short TR/TE	Fat bright, CSF dark	Anatomy, post-gadolinium, subacute blood, fat
T2w	Long TR/TE	Fluid bright	Oedema, most pathology, CSF spaces
FLAIR	IR nulling CSF + T2	Lesions bright, CSF dark	Periventricular/juxtacortical lesions (MS, small-vessel disease)
STIR	IR nulling fat + T2	Oedema bright, fat dark	Marrow oedema, MSK, whole-body screening
DWI/ADC	Diffusion gradients (b-values)	Restriction: DWI bright / ADC dark	Acute stroke, abscess, tumour cellularity
SWI T2*w	/ Long-TE phase mask	GRE + Blood products, iron, veins black	Microbleeds, haemorrhage, trauma, vascular malformations

Module 3 take-home. Contrast is not colour; it is a deliberate choice of which physical property to amplify. The same brain yields anatomy (T1w), pathology (T2w), lesion visibility (FLAIR), fat-free oedema (STIR), cellular water motion (DWI/ADC) and blood products (SWI) — all from the same protons, asked different questions.

MODULE 4

Pulse Sequences: The Engine Underneath

Learning objectives

- Read a basic pulse-sequence timing diagram (RF, gradients, signal).
- Explain how SE, TSE/FSE, GRE, EPI and IR generate echoes and fill k-space.
- Match each sequence family to its strengths, speed and characteristic artefacts.

A **pulse sequence** is the exact timetable of RF pulses and gradient switches that produces an image. Module 3 told you *what* contrast each weighting shows; this module tells you *how* the machine manufactures it. One concept unlocks the whole family tree: **how the echo is formed**.

4.1 Spin echo (SE): the honest rewind

After a 90° pulse, spins dephase by both true T2 and reversible T2' effects. SE inserts a 180° pulse at TE/2, which flips the dephasing spins so the fast ones are now behind and the slow ones ahead — they re-converge into an **echo** at TE. Field inhomogeneity is rewound; only true tissue T2 remains. SE is therefore the reference standard for T2 contrast — but one excitation fills one k-space line, so it is slow.

Picture it. Runners of different speeds start together and spread out around a track. At the halfway signal (the 180° pulse) everyone turns and runs back the way they came. Fast runners have further to return, slow runners less — and everyone crosses the start line at the same instant. That reunion is the echo.

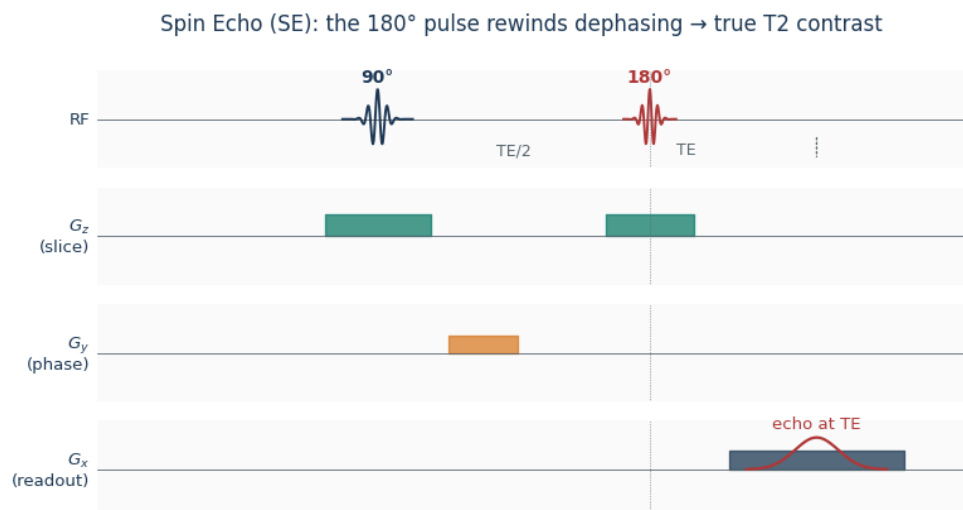


Figure 4.1. SE timing: 90° excitation with slice-select gradient, phase-encoding blip, 180° refocusing at $TE/2$, echo collected at TE during the readout gradient.

4.2 Turbo / fast spin echo (TSE/FSE): the production line

TSE applies a whole train of 180° pulses after one excitation, harvesting several echoes — and several k-space lines — per TR. The **echo train length (ETL)** sets the speed-up: ETL 8 means roughly $8\times$ faster than SE. The cost is subtle: echoes collected at different effective TE s are mixed into one image, giving mild blurring and altered fat contrast (fat stays bright on TSE T2 — a common trap when comparing with SE T2). Virtually all clinical T2w imaging today is TSE/FSE.

Turbo/Fast Spin Echo (TSE/FSE): many 180° refocusing pulses fill several k-space lines per TR

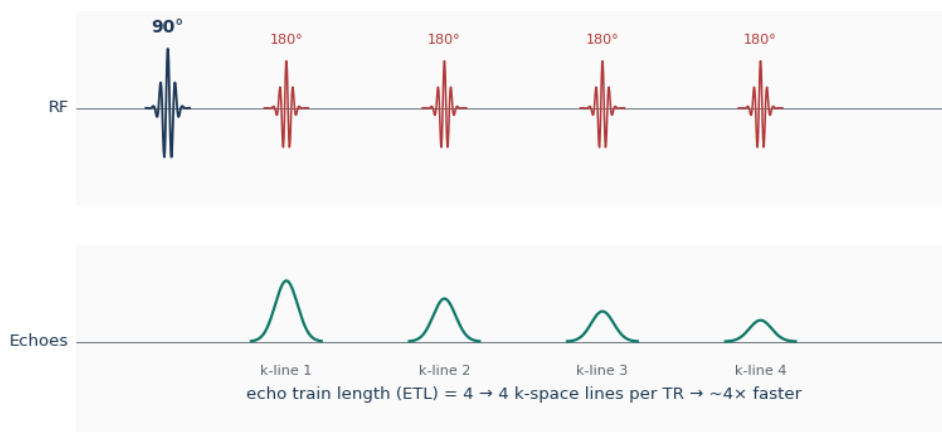


Figure 4.2. TSE/FSE: multiple 180° refocusing pulses after a single 90° excitation, each echo filling a different k-space line.

4.3 Gradient echo (GRE): speed without the rewind

GRE abandons the 180° pulse entirely. A readout gradient is first applied in reverse (dephase lobe), then flipped (rephase lobe): spins swept apart by the gradient are swept back together into an echo. Nothing rewinds the T_2' effects from field inhomogeneity, blood products or iron — so GRE is **T_2^* -weighted**, fast (small flip angles, short TR), 3D-capable, and exquisitely sensitive to susceptibility. Haemorrhage, calcification and air–tissue interfaces bloom dark. The same sensitivity that makes GRE the basis of SWI and BOLD fMRI also makes it vulnerable to distortion near sinuses, mastoids and implants.

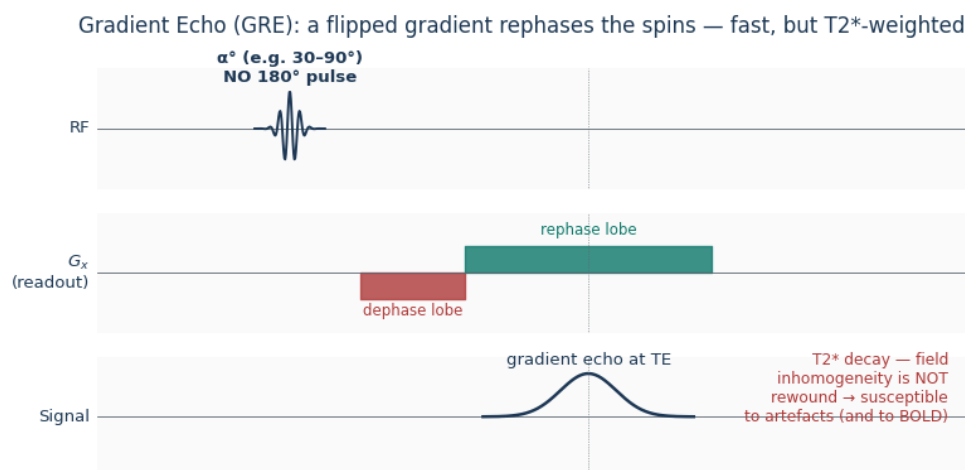


Figure 4.3. GRE timing: one RF pulse, a dephase/rephase gradient pair forming the echo, no 180° pulse — so T_2^* decay proceeds unchecked.

4.4 Echo planar imaging (EPI): the speed demon

EPI asks the extreme question: how much of k-space can be filled after *one* excitation? Answer — all of it. The readout gradient oscillates rapidly back and forth, sweeping a zigzag trajectory through k-space in a single shot lasting tens of milliseconds. A whole slice per shot, a whole brain per second or two: this is what makes DWI, DTI and fMRI practical. The price is that T_2^* decay continues *during* the readout, so EPI images suffer geometric distortion (worst near air–tissue interfaces), signal dropout, ghosting and blurring. Every fMRI preprocessing pipeline (distortion correction, motion correction) exists because of these artefacts.

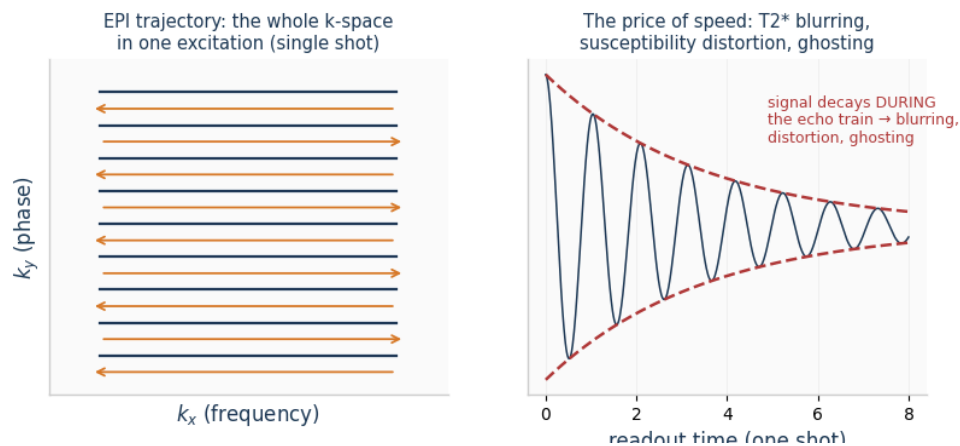


Figure 4.4. Left: the single-shot EPI trajectory through k-space. Right: signal decays during the long readout — the origin of EPI's blurring, distortion and ghosting.

4.5 Inversion recovery (IR): contrast by subtraction

IR prepends a 180° *inversion* pulse before the imaging sequence, then waits a chosen inversion time (TI) while magnetizations recover through zero (Figure 3.4). Image at a tissue's zero-crossing and that tissue is erased: FLAIR (TI ≈ 2900 ms) erases CSF, STIR (TI ≈ 180 ms) erases fat. A third family member, **MP-RAGE/MPRAGE**, combines an inversion preparation with fast 3D spoiled GRE to produce the high-resolution T1w anatomical volumes that neuroimaging pipelines (FreeSurfer, SPM, FSL) expect as their structural input.

Table 4.1. Sequence families at a glance

Family	Echo formed by	k-space per excitation	Weighting	Strength	Weakness / artefact
SE	180° RF pulse	1 line	T1, T2, PD	Robust, true T2	Slow
TSE/FSE	Train of 180° pulses	ETL lines	T2 (routine)	Fast clinical T2	Blurring; fat stays bright
GRE	Gradient reversal	1 line	T2*, T1 (spoiled)	Fast, 3D, susceptibility	Distortion, signal voids
EPI	Oscillating gradient	Whole grid (single shot)	T2*, DWI	Fastest; enables DWI/fMRI	Distortion, ghosting, dropout
IR	180° prep + SE/GRE readout	varies	T1-controlled	Nulls one tissue (FLAIR, STIR)	Long; TI must match T1

Module 4 take-home. SE rewinds with RF and is honest but slow; TSE rewinds many times and is the clinical workhorse; GRE rewinds with gradients and trades fidelity for speed and susceptibility sensitivity; EPI fills everything at once and pays with artefacts; IR deletes a chosen tissue. "Sequence" and "weighting" are not synonyms — a weighting is a contrast goal, a sequence is the machinery that reaches it.

MODULE 5

fMRI Physics: Imaging the Working Brain

Learning objectives

- Explain the BOLD mechanism from neuron to pixel.
- Describe the haemodynamic response function and its consequences for experimental design.
- Explain why fMRI is built on EPI and what its characteristic artefacts are.

5.1 BOLD: blood as an endogenous contrast agent

Functional MRI measures no neurons directly. It exploits a lucky piece of physiology and one piece of physics you already know — T_2^* .

The physiology: when a brain region becomes active, local blood flow increases *far more* than oxygen consumption does — an overshoot called neurovascular coupling. The physics: haemoglobin's magnetism depends on whether it carries oxygen. **Deoxyhaemoglobin is paramagnetic** (it disturbs the local field and accelerates T_2^* decay); **oxyhaemoglobin is diamagnetic** (near-invisible to the field). When activity flushes a region with oxygenated blood, deoxyhaemoglobin is diluted, the local field calms, T_2^* lengthens slightly, and the T_2^* -weighted signal rises by a few percent. That faint rise is the **BOLD** signal — blood-oxygen-level-dependent contrast.

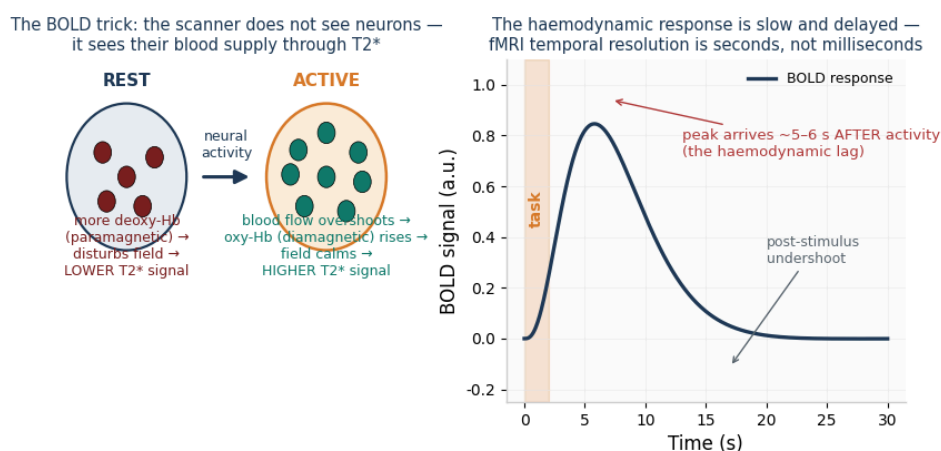


Figure 5.1. Left: rest vs activation — oxygen-rich blood displaces paramagnetic deoxyhaemoglobin, calming local field disturbances and raising T_2^* signal. Right: the response peaks 5–6 s after the neural event and is followed by an undershoot.

Keep the chain honest. BOLD = neural activity → neurovascular coupling → oxy/deoxy ratio → local field uniformity → T2* → GRE-EPI signal. fMRI conclusions inherit *every* link in that chain — including vascular disease, caffeine, breathing and scanner artefacts, none of which are neural.

5.2 The haemodynamic response and its lag

The blood-flow response is slow: after a brief neural event, BOLD signal rises over ≈ 2 s, peaks at 5–6 s, and settles over 15–20 s, often with a post-stimulus undershoot. This **haemodynamic response function (HRF)** acts like a smearing filter between neural activity and measurement. Three practical consequences:

- **Temporal resolution is seconds**, no matter how fast the scanner is — fMRI cannot order millisecond-scale neural events.
- **Designs must be convolved with the HRF**: analysis asks which voxels' time series match the task-timing pattern *after* blurring by the HRF (Figure 5.2).
- **Events too close together overlap**: block designs (20–40 s ON/OFF) maximize detectability; event-related designs trade power for trial-level detail.

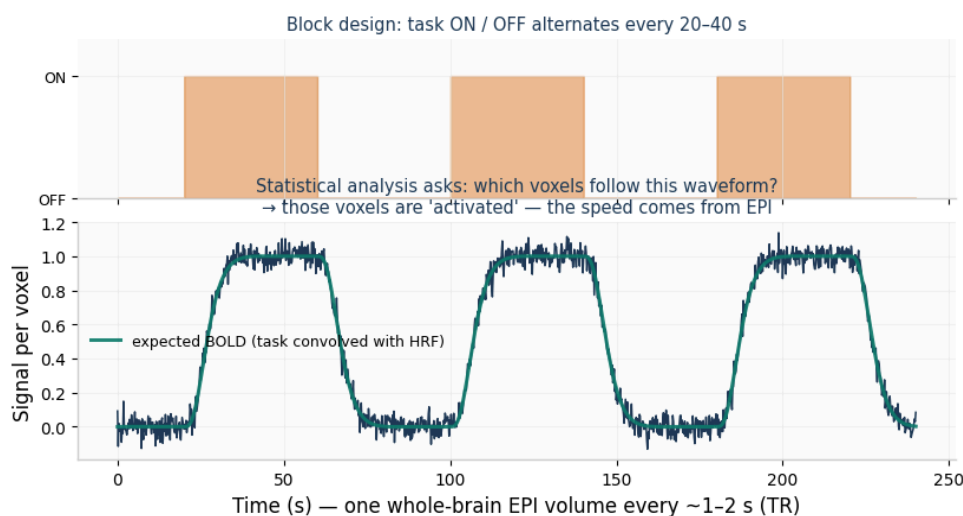


Figure 5.2. A block design (top) and the voxel time series it predicts after convolution with the HRF (bottom). Statistical parametric mapping is, at heart, a correlation between each voxel's measured signal and this expected waveform.

5.3 Why fMRI needs EPI

BOLD changes are a few percent of signal, so fMRI must sample the whole brain every 1–2 s for several minutes and squeeze statistics out of the time series. Only EPI (Module 4.4) is fast enough: single-shot T2*-weighted GRE-EPI is the default fMRI acquisition. Every property of fMRI data — the distortion, the dropout near the orbitofrontal cortex and temporal poles, the sensitivity to head motion — is an EPI property wearing a neuroscience costume.

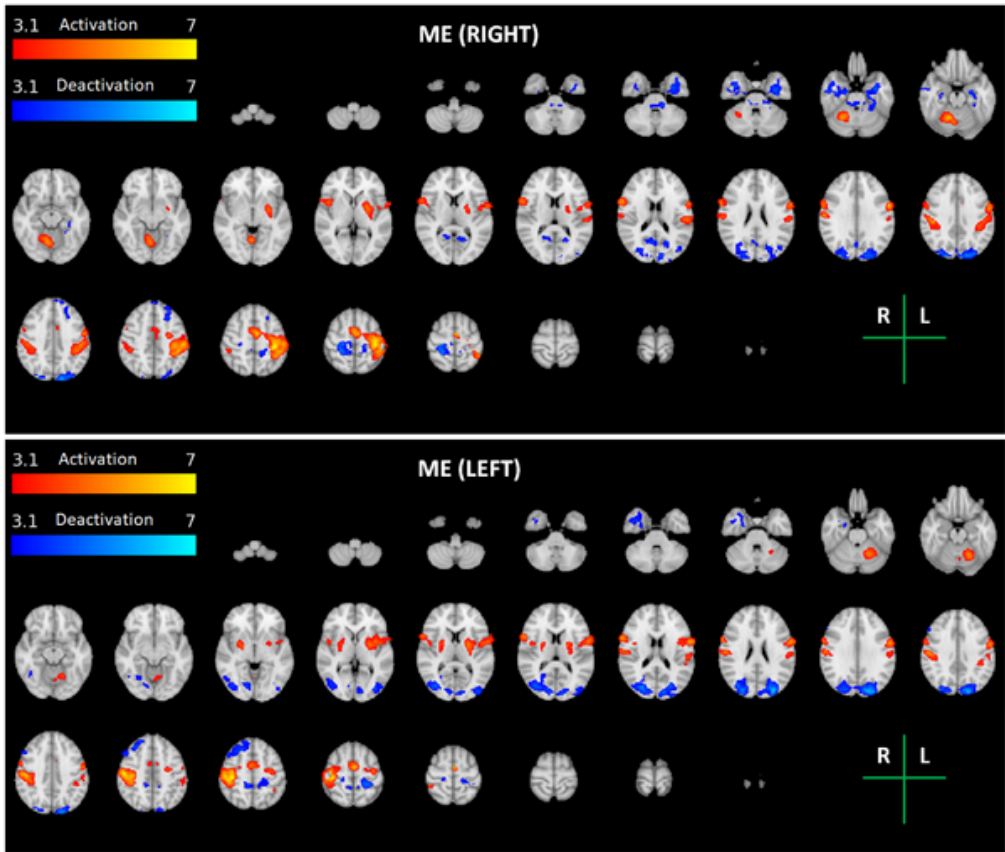


Figure 5.3. BOLD activation maps (warm colours = activation, cool = deactivation) overlaid on structural MRI during a motor task — the end product of the entire chain in this module. Image: PLOS ONE, brain activation during a VR-based motor imagery/observation task (CC BY).

5.4 Artefacts every fMRI scientist must know

Table 5.1. The fMRI artefact shortlist

Artefact	Physical origin	Looks like	Standard mitigation
Susceptibility dropout	T2* destroyed near air–tissue interfaces	Signal voids: orbitofrontal, anterior temporal	Shorter TE, shimming, distortion correction
Geometric distortion	B ₀ inhomogeneity warps EPI readout	Stretched/compressed frontal & temporal lobes	Field-map or reverse-polarity (topup) correction
Head motion	k-space inconsistencies between volumes	Edge ghosts, spurious "activation", group differences	Realignment, motion regressors, scrubbing
Physiological noise	Cardiac/respiratory cycles alias into BOLD	Fluctuations at vessel/CSF locations	RETROICOR, ICA-based denoising
Scanner drift	Gradual instabilities hardware	Slow trends mistaken for sustained activity	High-pass detrending filtering,

Module 5 take-home. fMRI = T2* + neurovascular overshoot + EPI speed + statistics. The scanner sees blood, not thoughts; the HRF smears seconds; EPI supplies the speed and the artefacts. Understanding this module is the difference between analysing brain function and analysing a magnetic weather report.

Summary — The Chain of MRI

$\mathbf{B}_0$ aligns the spins $\rightarrow \omega = \gamma \mathbf{B}_0$ sets the frequency $\rightarrow \mathbf{B}_1$ tips M into the transverse plane $\rightarrow \mathbf{T1/T2/T2^*}$ give tissues their signatures $\rightarrow$ **gradients** encode position as frequency and phase $\rightarrow$ **k-space** accumulates the coded signal $\rightarrow$ **FFT** reconstructs the image $\rightarrow$ **TR/TE/TI/b** choose the contrast $\rightarrow$ **the sequence** (SE / TSE / GRE / EPI / IR) is the timetable that makes it happen $\rightarrow$ **BOLD fMRI** exploits T2* and blood flow to watch the brain work.

Every term in that chain is now in your vocabulary. When a dataset behaves strangely — a model that fails on a new scanner, a lesion that appears on DWI but not ADC, an activation cluster beside the sinuses — walk the chain and ask at which link physics, not biology, is speaking.

Cheat Sheet & Glossary

Table 6.1. One-line answers to the questions fellows actually ask

Question	One-line answer
Why hydrogen?	Most abundant nucleus in the body and the loudest ($\gamma = 42.58 \text{ MHz/T}$).
What does B_0 do?	Aligns spins, creates net M, sets precession frequency. Always on.
What does B_1 do?	Tuned RF pulse that tips M into the measurable plane (flip angle).
What do gradients do?	Make frequency/phase position-dependent: slice (z), phase (y), frequency (x).
T1 vs T2 in one line?	T1 = regrowth along the field (energy out); T2 = loss of sync across it (coherence gone). $T2 \leq T1$.
What is T2*?	T2 plus field-inhomogeneity decay; what GRE and EPI actually see; BOLD's currency.
What is k-space?	The raw frequency/phase data matrix; image = its inverse FFT. Centre = contrast, edge = detail.
Short TR + short TE?	T1-weighted (anatomy; fat bright, fluid dark).
Long TR + long TE?	T2-weighted (pathology; fluid bright).
How to kill CSF signal?	FLAIR: invert, image at TI $\approx 2900 \text{ ms}$ (CSF zero-crossing).
How to kill fat signal?	STIR: invert, image at TI $\approx 180 \text{ ms}$ (fat zero-crossing).
DWI bright means stroke?	Only if ADC is dark. DWI bright + ADC bright = T2 shine-through.
What is SWI for?	Blood products, iron, microbleeds, veins — anything that disturbs the local field.
Why is EPI everywhere?	Fills all of k-space in one shot; the only readout fast enough for DWI and fMRI.
What does fMRI measure?	Not neurons — the BOLD blood-flow overshoot via T2*, delayed $\sim 5\text{--}6 \text{ s}$.

Glossary (selected)

Term	Meaning
B_0	Main static magnetic field (1.5–7 T), produced by the superconducting magnet.
B_1	Oscillating radiofrequency field used to excite spins.
Flip angle (α)	Angle by which RF tips the net magnetization away from B_0 .
Larmor frequency	Spin precession frequency, $\omega = \gamma B_0$; the resonance condition for RF.
Gyromagnetic ratio (γ)	Nucleus-specific constant linking field strength to precession frequency.
TR / TE / TI	Repetition time / echo time / inversion time — the timing dials of contrast.

b-value	Strength of diffusion weighting in DWI (s/mm^2).
ADC	Apparent diffusion coefficient; quantitative map of water mobility computed from multi-b DWI.
FID	Free induction decay — the raw decaying signal after one excitation.
SAR	Specific absorption rate — RF power deposited in tissue (W/kg); safety-limited.
BOLD	Blood-oxygen-level-dependent contrast — the basis of fMRI.
HRF	Haemodynamic response function — the slow BOLD waveform following neural activity.
Partial volume	Averaging of multiple tissues within one voxel, blurring borders and small lesions.
SNR	Signal-to-noise ratio; rises with field strength, voxel size and averaging.

Further Reading

- McRobbie, D. W., Moore, E. A., Graves, M. J., & Prince, M. R. (2017). *MRI from Picture to Proton* (3rd ed.). Cambridge University Press. — The clearest intuition-builder; start here.
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- Bernstein, M. A., King, K. F., & Zhou, X. J. (2004). *Handbook of MRI Pulse Sequences*. Elsevier Academic Press. — For Module 4 in full depth.
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- Huettel, S. A., Song, A. W., & McCarthy, G. (2014). *Functional Magnetic Resonance Imaging* (3rd ed.). Sinauer Associates. — The standard fMRI text for Module 5.
- Prince, J. L., & Links, J. M. (2014). *Medical Imaging Signals and Systems* (2nd ed.). Pearson. — The signal-processing foundation.

Figure Credits

- Figures 1.1–1.3, 2.1–2.6, 3.1, 3.4, 3.5, 3.7, 4.1–4.4, 5.1–5.2 were generated specifically for these notes from the governing equations (Larmor equation, Bloch-equation relaxation, Fourier transform of a standard test phantom, canonical haemodynamic response model).
- Figure 3.2: Wikimedia Commons, file "T1t2PD.jpg" (commons.wikimedia.org/wiki/File:T1t2PD.jpg).
- Figure 3.3: Case Western Reserve University, School of Medicine (Neurology), "MRI Basics" ([case.edu/med/neurology/NR/MRI Basics.htm](http://case.edu/med/neurology/NR/MRI%20Basics.htm)).
- Figure 3.6: Stroke Manual, "MR-DWI in the acute stroke diagnosis" (stroke-manual.com/mri-dwi-in-stroke-diagnosis).
- Figure 3.8: *Stroke* / American Heart Association Journals, figure from "Susceptibility-Weighted Imaging is More Reliable Than T2*-Weighted Gradient-Recalled Echo MRI for Detecting Microbleeds" (ahajournals.org/doi/10.1161/STROKEAHA.113.002267).
- Figure 5.3: PLOS ONE, "Brain activation by a VR-based motor imagery and observation task: an fMRI study" (journals.plos.org/plosone/article?id=10.1371/journal.pone.0291528), published under a Creative Commons Attribution (CC BY) licence.
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